



Effects of feeding silage inoculated with *Lactococcus lactis* and *Lentilactobacillus buchneri* on performance and nutrient utilization in transition dairy cows

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ABSTRACT

Two experiments were conducted to assess the effect of including a mixed microbial inoculant on (1) the fermentation and nutritional value of silage in laboratory-scale silos and (2) the effects of feeding an inoculated silage-based TMR on the performance, nutrient digestibility, and ruminal fermentation in transition dairy cows from –21 d to +42 DIM. In experiment 1, harvested fresh corn forage was inoculated either with (1) double-distilled water (CON-L) or (2) mixed microbial inoculant with 1.5×10^5 cfu/g of fresh forage of *Lactococcus lactis* O224 and *Lentilactobacillus buchneri* LB1819 at a rate of 30 mL/kg of fresh forage (INO-L). Adding mixed microbial inoculants under laboratory conditions had no effect on the chemical composition of the corn silage; however, the fermentation parameters and aerobic stability were improved after 90 d of ensiling. Acetate and 1,2-propanediol increased for INO-L after 90 d of ensiling (1.72% vs. 1.17% DM and 0.47% vs. 0.01% DM, respectively). Moreover, INO-L had lower yeast counts than CON-L after 90 d of ensiling (2.07 vs. 3.63 log cfu/g). Dry matter loss was lower for INO-L than CON-L (1.06% vs. 3.74%). Lastly, the INO-L treatment had greater aerobic stability than CON-L (230.8 vs. 95.1 h). For experiment 2, fresh corn and ryegrass were ensiled in Ag-Bags with (1) the control treatment, double-distilled water, or (2) the inoculated treatment, a mixed microbial inoculant with 1.5×10^5 cfu/g of fresh forage of *Lactococcus lactis* O224 and *L. buchneri* LB1819 at a rate of 30 mL/kg of fresh forage, and mixed in the diets of prepartum and postpartum dairy cows. Fifty-four pregnant cows (245 ± 3 d of gestation; 768 ± 76 kg BW) were fed a TMR with

control or inoculated corn and ryegrass silage (54% and 9.3% of diet DM, and 43% and 6.7% of diet DM during the prepartum and postpartum phases, respectively). Feeding TMR with inoculated silages did not affect the DMI and ruminal fermentation of prepartum cows. However, DMI tended to increase at ~4 wk postpartum for cows fed the TMR with inoculated silage. Moreover, 3.5% FCM and ECM yields increased for postpartum cows fed the TMR with inoculated silages (44.9 vs. 42.0 kg/d and 44.5 vs. 41.9 kg/d, respectively). Milk fat yield increased for cows fed the TMR with inoculated silages compared with those fed the TMR with control silages (1.62 vs. 1.50 kg/d). No treatment effects were observed on the total-tract apparent nutrient digestibility of prepartum cows. However, we observed a tendency to increase NDF digestibility for postpartum cows fed the TMR with inoculated silages compared with those fed the TMR with the control silages (45.2% vs. 42.0%). Prepartum cows fed inoculated silages had greater ammonia-N concentrations than those fed the TMR with the control silages (5.47 vs. 4.18 mM). No treatment differences were observed in ruminal fermentation during the postpartum phase. In conclusion, microbial additives improved the aerobic stability of preserved forages at the feed-out phase, and feeding inoculated silage-based TMR had no effects for prepartum cows but enhanced the production performance of lactating dairy cows in early lactation.

Key words: microbial inoculants, aerobic stability, transition cows

INTRODUCTION

Corn silage has been one of the most common forage sources for lactating dairy cows in the United States (Ferraretto et al., 2018). Based on a recent estimate (USDA-NASS, 2021), almost 2.7 million hectares of

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

corn were harvested in the United States, and 130 million tonnes were used to produce corn silage, yielding ~8.1 t per hectare of arable land. Corn silage is a good source of starch and effective fiber required to support greater levels of milk production, and effective ensiling is essential to preserve the nutritional value of the harvested corn forage. Similarly, cool season grasses such as perennial ryegrass are commonly used to feed US dairy herds. Ryegrass is characterized by greater NDF and lower proportions of undigested NDF (**uNDF**) than alfalfa (Hoffman et al., 1993). Bernard et al. (2002) observed that the partial or total replacement of corn silage with ryegrass silage improved milk yield (**MY**) and milk components (i.e., fat and protein) of lactating dairy cows.

Over the years, research has focused on evaluating the effects of additives for improving fermentation characteristics and increasing the aerobic stability of preserved forages. The inclusion of microbial additives is a common practice to ensure good preservation by reducing the influence of epiphytic bacteria on ensiled forages (Arriola, 2010) and maintaining aerobic stability for prolonged periods after aerobic exposure. Homolactic bacteria (e.g., *Lactococcus lactis*) ferment hexoses, primarily glucose, into lactic acid (Muck et al., 2018), decreasing the pH faster during the early fermentation phase of ensiling (Muck, 1988), thereby inhibiting the activity of epiphytic bacteria present in the harvested forage and inducing faster and better fermentation. Obligate heterolactic bacteria (e.g., *Lentilactobacillus buchneri*; Adesogan and Arriola, 2020) ferment the lactic acid produced during anaerobic fermentation into acetic acid and 1,2-propanediol (Oude Elferink et al., 2001). Aerobic stability is increased during the feed-out phase because greater acetic acid has the potential to inhibit yeast growth, which is responsible for initiating aerobic spoilage (Muck et al., 2018). Muck et al. (2018) emphasized the importance of combining homo- and heterofermentative bacteria to maintain fermentation characteristics while improving the aerobic stability of ensiled forages.

Nutritional management during the transition phase in dairy cows is critical due to sudden changes in energy requirements before and after calving, which may lead to negative energy balance status during early lactation in response to reduced DMI. Improper nutritional management during the transition period may result in greater incidences of metabolic disorders and infectious diseases due to immunosuppression (Drackley et al., 2001; Mulligan and Doherty, 2008). Although extensive research has been conducted on the nutritional management of transition dairy cows, we still lack data exploring the effects of forage quality in transition dairy cows, including changes in ruminal fermentation, blood parameters, and performance.

We hypothesized that using a mixed microbial inoculant containing *L. lactis* and *L. buchneri* would reduce pH at a faster rate during the ensiling phase and improve the aerobic stability of corn silage ensiled under laboratory conditions. In addition, we hypothesized that feeding a TMR constituted with inoculated corn and ryegrass silage would increase prepartum and postpartum DMI as well as the performance of lactating dairy cows during early lactation. Therefore, the objectives of this experiment were to assess the effect of using microbial inoculants on the fermentation parameters and aerobic stability of corn forage ensiled under laboratory conditions and to evaluate the impact of feeding TMR with inoculated corn and ryegrass silage on performance and ruminal fermentation in transition dairy cows.

MATERIALS AND METHODS

The University of Florida Institutional Animal Care and Use Committee approved all animal care and experimental procedures for this experiment (protocol #202011194).

Experiment 1

Forage and Treatment. Corn (P1847VYHR, Pioneer, Johnston, IA) was grown at the Dairy Research Unit at the University of Florida and harvested at 40.7% of DM with a Claas Jaguar 850 forage harvester equipped with a kernel processor (1.5-mm roll gap settings, 30% differential speed, Claas of America, Columbus, IN). The forage was chopped at a theoretical length of 19 mm. Corn forage was harvested from 5 different locations (blocks) across the field, and forage from each block was split into two 50-kg piles and placed on separate plastic sheets. The following treatments were randomly applied to each block: (1) double-distilled water at 30 mL/kg (**CON-L**), and (2) inoculation with SiloSolve FC (Novonesis, Lyngby, Denmark) containing 1.5×10^5 cfu/g of *L. lactis* O224 and *L. buchneri* LB1819 and applied at 30 mL/kg of forage (**INO-L**). Both CON-L and INO-L treatments had 5 replicates, representing the different locations in the experimental plot. Each replicate was placed into 20-L capacity buckets, filled with ~11 kg of fresh chopped corn, sealed, and stored for 30 and 90 d to measure nutritional composition, fermentation characteristics, and microbial counts. Corn forage was also ensiled for 7, 14, 30, and 90 d to measure aerobic stability and VFA using 4-L buckets filled with ~2.5 kg of fresh forage. All buckets were weighed at the beginning and end of ensiling to account for DM losses.

Chemical Analysis. Before ensiling (d 0) and after 30 and 90 d of ensiling, all replicates for each treatment were opened, and a subsample was taken, dried at 60°C

for 48 h, and ground to pass through a 1-mm screen using a Wiley mill (A.H. Thomas, Philadelphia, PA). Concentrations of NDF and ADF were determined sequentially (Van Soest et al., 1991) using an Ankom 200 fiber analyzer (Ankom Technologies, Macedon, NY) with heat-stable α -amylase and sodium sulfite. Volatile fatty acids, ammonia-N ($\text{NH}_3\text{-N}$), and CP analyses were performed at the Dairyland Laboratories (Arcadia, WI). Crude protein was determined using the AOAC International (1995) combustion method, and VFA were determined using HPLC, as described previously (Canale et al., 1984). Ammonia-N was determined by distillation following the recommended methods of manure analysis (Peters et al., 2003). Starch was determined based on the method proposed by Hall (2015). Organic matter and ash contents were determined using AOAC International (2012) method 942.05.

Microbial Counts. Silage extract was prepared by mixing 20 g of fresh silage with 180 mL of 0.1% peptone water, blended for 1 min in a Stomacher (Lab-Blender 400, Tekmar Company, Cincinnati, OH), and filtered using 2 layers of cheesecloth. One portion of the filtrate was used to measure silage pH using a pH meter (Corning model 12, Corning Scientific Instruments, Medfield, MA). Serial dilutions were prepared on the remaining filtrate, and yeast and mold counts were determined using the pour plate method with malt extract agar (Thermo Fisher Scientific-Oxoid CM0059B) and incubated at 32°C for 72 h. Lactic acid at an 85% concentration was added to inhibit the growth of bacteria (0.5% vol/vol). Colony-forming units of molds and yeasts were enumerated separately according to their macro-morphological characteristics on plates that showed 1 to 100 cfu.

Aerobic Stability. Forages were exposed aerobically and allowed to deteriorate at room temperature in 20-L plastic buckets protected from dust contamination with 2 layers of cheesecloth. Aerobic stability was measured from forages ensiled at 7, 14, 30, and 90 d using ~2.50 kg of fresh silage from each bucket. Onset HOBO data loggers (Onset Computer Corporation, Bourne, MA) were placed in the geometric center of each bucket and programmed to measure the temperature every 30 min. Two loggers were kept at room temperature and programmed to measure the temperature every 30 min. Buckets were aerobically exposed for 14 d, then the loggers were removed and all the stored data were read. Aerobic stability was defined as the number of hours the silage remained stable before the temperature increased by more than 2°C above room temperature.

Experiment 2

Forages and Treatments. Corn (P1847VYHR, Pioneer, Johnston, IA) was planted at the Dairy Unit at the Uni-

versity of Florida (Gainesville, FL) on January 27, 2020, and harvested on June 25, 2020, at 40.7% of DM and chopped at a theoretical length of 19 mm using a Claas Jaguar 850 forage harvester equipped with a kernel processor (1.5-mm roll gap, 30% differential speed, Claas of America, Columbus, IN). Annual ryegrass (Earlyploid, Mayo Fertilizer Inc., Lake City, FL) was planted at the Dairy Unit at the University of Florida on December 3, 2020, and harvested on March 3, 2021, at 25% of DM and chopped at a theoretical length of 19 mm using the same harvester used for corn. Chopped annual ryegrass was wilted for 96 h before ensiling, aiming for 55% of DM. The treatments were (1) control: application of double-distilled water (**CON-AC** for corn silage and **CON-AR** for ryegrass silage), and (2) inoculated: application of SiloSolve FC (Novonesis, Lyngby, Denmark) containing 1.5×10^5 cfu/g of *L. lactis* O224 and *L. buchneri* LB1819 and applied at 30 mL/kg of forage (**INO-AC** for corn silage and **INO-AR** for ryegrass silage). Inoculants were applied during harvest, and to ensure uniformity in the forages used for both treatments, the inoculant was applied using a sprayer mounted on the harvester on alternate 8-row-wide swaths of forage, and the respective swaths of forages were alternately packed into Ag-Bag silos (Ag-Bag, Miller-St. Nazianz Inc., St. Nazianz, WI) with a Versa 1012 bagger (Versa Corp., Astoria, OR). Corn and ryegrass fields were divided into blocks such that forage from the first windrow in each block was packed into the first replicate used for the control treatment, and forage from the second windrow was packed into the first replicate used for the inoculated treatment. To maintain separation between treatments, all control replicates were harvested and packed first, followed by the inoculated swaths. This sequencing was intentional to eliminate the possibility of cross-contamination of the controls with inoculant. Two 5-ton Ag-Bags were made for each treatment for corn and ryegrass (4 bags per forage). Corn was ensiled for 395 d, and ryegrass was ensiled for 98 d. Representative samples were collected 3 times per week throughout the experiment. Dry matter was calculated to adjust the diet of the cows. The remaining samples were dried at 60°C for 48 h, ground to pass a 1-mm sieve, and stored for analysis. A composite by week was made for the chemical composition of both corn and ryegrass. Subsequently, we analyzed 10 samples per Ag-Bag for corn and ryegrass. All chemical composition analyses for corn were performed using wet chemistry, and ryegrass samples were analyzed using the near-infrared reflectance spectroscopy method at Dairyland Laboratories (Arcadia, WI).

Cows, Housing, and Experimental Design. The experiment was performed at the University of Florida Dairy Unit (Gainesville, FL). Fifty-four multiparous dry cows (245 ± 3 d pregnant; 768 ± 76 kg of BW) were

enrolled based on the selection criteria of a healthy pregnancy with no history of disease in the 90 d preceding experiment enrollment. Cows were housed in a freestall experimental barn with individual sand beds and assigned to individual feeding gates (Calan Broadbent feeding system, American Calan Inc., Northwood, NH). The experimental pens had one fan every 6 m above the beds and water sprinklers above the feeding gates. Fans and water sprinklers were automatically activated when the thermostats reached 18°C. The experimental duration was from 21 d before the expected calving date until 42 DIM.

The experiment followed a randomized complete block design in which prepartum cows were blocked by their previous 305-d ECM yield. A total of 27 blocks were formed, with 2 cows per block, and treatments were randomly assigned within each block. The treatments were (1) a TMR composed of uninoculated corn and ryegrass silage (**CON**), and (2) a TMR composed of inoculated corn and ryegrass silage (**INO**). Prepartum and postpartum diets were formulated using the Agricultural Modeling and Training Systems ration formulation software (AMTS.Cattle.Professional based on the Cornell Net Carbohydrate and Protein System 6.5.5) to estimate nutrient requirements and calculate the supply of nutrients to experimental animals, given nutrient profile and feed digestibility. Diets were formulated for lactating dairy cows producing 38.5 kg/d of milk, with 3.70% milk fat and 3.10% milk true protein and an expected DMI of 21 kg/d.

Milk Production, Milk Composition, and Body Weight.

Cows were milked twice daily at approximately 1000 and 2200 h. Milk yield was recorded using the Afimilk system (Afilab Mini Laboratory, Afimilk Ltd., Kibbutz Afikim, Israel). Milk samples were collected from 4 consecutive milkings (morning and evening for 2 consecutive d) every week for the postpartum period (i.e., 6 weeks). Milk samples were collected in plastic vials and stored at 4°C with a preservative (bronopol-B2) prior to analysis at the Southeast Milk dairy laboratory (Bellevue, FL) for fat, true protein, MUN, lactose, and total solids using a Fourier transform spectrometer (model FTS 500, Bentley Instruments Inc., Chaska, MN), and SCC using a Bentley flow cytometer (model FCM 500, Bentley Instruments Inc., Chaska, MN). The milk components yield was adjusted for morning and afternoon MY. The ECM yield was calculated as $[(0.3246 \times \text{MY}) + (12.86 \times \text{milk fat yield}) + (7.04 \times \text{milk protein yield})]$, according to NRC (2001). The 3.5% FCM yield was calculated as $[(0.4324 \times \text{MY}) + (16.218 \times \text{milk fat yield})]$ according to NRC (2001). Statistical analysis was performed with weekly mean values obtained from the calculated daily values. Body weight was recorded weekly during the prepartum period using a walk-through scale (Tru-Test Inc., Mineral

Wells, TX), and ADG was calculated based on initial and final weight. During the postpartum period, BW was recorded twice daily after milking using an Afimilk scale (Afimilk Ltd., Kibbutz Afikim, Israel).

Intake and Nutrient Digestibility. Experimental animals were fed ad libitum once daily at around 0700 h, and the daily amount was adjusted for 5% refusals. Individual intake was recorded by weighing the feed offered and refusals daily before feeding. Total mixed ration and refusal samples were collected thrice weekly and dried in a forced-air oven at 60°C for 48 h for DM estimation and DMI calculations. Dried samples were stored at room temperature for further analysis.

Nutrient digestibility was estimated during the prepartum (3 and 1 wk before calving) and postpartum (2 and 6 wk after calving) periods using uNDF as an internal marker. Feed offered was sampled immediately after feeding, whereas spot fecal samples were collected from every animal during digestibility measurements (d 1: 0500, 0900, 1300, and 1700 h; d 2: 0600, 1000, 1400, and 1800 h; d 3: 0700, 1100, 1500, and 1900 h). Fecal and TMR samples were dried at 60°C for 48 h in a forced-air oven, ground to pass through a 1-mm screen using a Wiley mill and stored at room temperature for further analysis. To estimate uNDF in TMR and fecal samples, ~0.70 g of sample was weighed into Ankom F57 filter bags (Ankom Technology, Macedon, NY) in triplicate and incubated for 288 h in 2 ruminally cannulated lactating cows that were fed a corn silage-based TMR (47% corn silage, 34% NDF, 17.7% CP, and 28% starch). Bags were removed from the cows and placed into 20-L buckets filled with ice-cold water to stop fermentation immediately. Subsequently, they were rinsed with cold water to remove fiber particles and washed in a regular washing machine using a normal rinse cycle before analyzing for NDF. Fecal output (kg of DM/d) was calculated as follows: $\text{uNDF intake (kg/d)/uNDF\% feces}$, where $\text{uNDF intake (kg/d)} = [\text{uNDF\% TMR} \times \text{TMR offered (kg/d)}] - [\text{uNDF\% orts} \times \text{orts (kg/d)}]$. Total-tract apparent digestibility (%) was calculated for all nutrients analyzed as follows: $100 - \{[\text{nutrient\% feces} \times \text{fecal output (kg/d)}]/[\text{nutrient intake (kg/d)}]\}$, where $\text{nutrient intake (kg/d)} = [\text{nutrient\% TMR} \times \text{TMR offered (kg/d)}] - [\text{nutrient\% orts} \times \text{orts (kg/d)}]$ (Huhtanen et al., 1994).

Aerobic stability was measured on the TMR using the same methods as in experiment 1, using fresh samples taken immediately after feeding pre- and postpartum cows.

Total Mixed Ration Chemical Analysis. Concentrations of NDF and ADF were determined sequentially using the van Soest method (van Soest et al., 1991) with an Ankom 200 fiber analyzer (Ankom Technologies, Macedon, NY). Heat-stable α -amylase and sodium sulfite were used for the NDF analysis. Nitrogen concentration

was determined by rapid combustion using a macro elemental N analyzer (Vario Micro CN, Elementar Americas Inc., Mt. Laurel, NJ) according to the official method 992.15 (AOAC International, 1995), and CP values were calculated by multiplying the N concentration value by 6.25. The ether extract was analyzed using a fat analyzer (XT20, Ankom Technology; AOAC International, 2000, method 2003.05).

Blood Sampling and Analysis. Blood samples were collected from the coccygeal vein into evacuated tubes containing sodium heparin anticoagulant on d -21, -14, and -7 prepartum and on d 0, 1, 2, 3, 7, 14, 21, 28, 35, and 42 postpartum. All samples were collected between 2 and 3 h after feeding, except for the day of calving, during which blood was collected within 1 h after calving. Tubes were centrifuged at $2500 \times g$ for 20 min at 4°C and stored at -20°C until further analysis. Plasma samples were analyzed for glucose (Glucose Oxidase Liquid reagent set, TECO Diagnostics Inc.), nonesterified fatty acids (NEFA; NEFA-C kit; Wako Diagnostics Inc.), and plasma urea nitrogen (PUN) concentrations (Arbor Assays, Ann Arbor, MI).

Ruminal Fermentation Characteristics and Rumination. Rumen fluid samples were collected using a stomach tube every week during the prepartum period (i.e., -3, -2, and -1 week) and during wk 2 and 6 in the postpartum period, approximately 2 h after feeding. A stomach tube was introduced through the mouth, and the fluid was pumped out of the rumen into a 1-L vessel. The first 200 mL were discarded to avoid saliva contamination, and the following 200 mL were filtered through 4 layers of cheesecloth. Ruminal pH was determined immediately after collection using a pH meter. The filtered samples (10 mL) were transferred to 15-mL tubes and was acidified with 0.1 mL of 50% H_2SO_4 solution (vol/vol) and centrifuged at $7,000 \times g$ for 15 min at 4°C and stored at -20°C until further analysis. The first sample from this supernatant was stored at -20°C for further $\text{NH}_3\text{-N}$ analysis (Technicon Auto Analyzer, Technicon), and a second sample was filtered using 0.22- μm filters and transferred to vials for VFA analysis using a Merck Hitachi Elite LaChrome HPLC system (Hitachi, L2400), a UV detector, and an Aminex HPX-87H column (Bio-Rad Laboratories). This column was used isocratically with a 0.015 M H_2SO_4 mobile phase and a 0.7 mL/min flow rate at 45°C .

All experimental cows received a neck collar and were registered in the software HealthyCow24 (Allflex SCR, Hi Tag, SCR Engineers Ltd., Netanya, Israel) for electronic monitoring of rumination and activity. Data on rumination time and activity index were recorded in minutes at 2-h intervals (total time in minutes spent ruminating in every 2-h interval across 24 h).

Urine Sampling and Analysis. Nitrogen balance was calculated using the N intake and the fecal, urine, and milk N output. Spot urine samples were collected twice daily at 0600 h and 1400 h for 2 consecutive days every week (i.e., -3, -2, and -1 week) in the prepartum period and during wk 2 and 6 in the postpartum period. Samples were filtered through 2 layers of cheesecloth, and 10 mL of urine per collection was added to a bottle containing 160 mL of 0.036 N H_2SO_4 and stored at -20°C for further analysis of uric acid, allantoin, creatinine, urinary urea N, and urine N. The uric acid concentration was determined using the Infinity Uric Acid Liquid Stable Reagent from Thermo Fisher Scientific Inc. (Middletown, VA), whereas allantoin concentration was measured using a colorimetric assay according to Chen and Gomes (1992). Creatinine and urea N concentrations were measured using the colorimetric detection kit from Arbor Assays (Ann Arbor, MI). Urine N was determined by rapid combustion using a macro elemental N analyzer (Vario Micro CN, Elementar Americas Inc., Mt. Laurel, NJ) based on AOAC International (1995) official method 992.15. Average daily urine output was estimated using creatinine concentration as a marker equivalent to 24 mg/kg BW per day (Chizzotti et al., 2008). Total purine derivatives (PD) were calculated as the sum of uric acid and allantoin excreted in the urine. Absorbed purines (AP) were estimated from the excretion of PD ($\text{PD} = 0.85 \times \text{AP} + 0.512 \times \text{BW}^{0.75}$) according to Cobianchi et al. (2012). Microbial N flow (g/d) was estimated using daily absorption of microbial purines as (microbial purine absorption $\times 70$)/(0.116 $\times 0.83 \times 1,000$) according to Chen and Gomes (1992). Only 7 blocks were used for N balance calculations.

Statistical Analysis

Experiment 1. Data were analyzed as a randomized complete block design (RCBD) using the GLIMMIX procedure of SAS (version 9.4, SAS/STAT, SAS Institute Inc., Cary, NC). The statistical model included the fixed effects of inoculation, ensiling duration, and their interaction, whereas block was used as a random factor. Data were tested for distribution of the residuals, and non-normally distributed residuals were subjected to Box-Cox transformation to meet the assumptions of normality before analyses. Significance was declared at $P \leq 0.05$, and a tendency at $P \leq 0.10$.

Experiment 2. Prepartum and postpartum data were analyzed separately using the MIXED or GLIMMIX procedures of SAS. Data were tested for the distribution of the residuals, and non-normally distributed residuals were subjected to Box-Cox transformation to meet the assumptions of normality before analyses. If the final

model after transformation remained the same and no changes in significance were observed, data analysis was carried out on the original dataset to avoid back transforming the SEM.

Milk production and DMI data were converted to weekly means for statistical analysis. Data were analyzed using an RCBD using the GLIMMIX procedure of SAS, and the covariance structure was selected based on the lowest Akaike information criterion. The statistical model for variables measured over time included treatment, time (week or days, depending on the variable), and 2-way interactions as fixed effects, whereas cows nested within a treatment were used as a random effect. Significance was declared at $P \leq 0.05$, and a tendency at $P \leq 0.10$. Data were presented as LSM, and SEM was reported. Interactions are reported only when significant.

RESULTS

Experiment 1

No treatment differences were observed for DM ($P = 0.85$), OM ($P = 0.89$), amylase-corrected NDF ($P = 0.23$), ADF ($P = 0.54$), starch ($P = 0.20$), and ash ($P = 0.89$; Table 1). However, water-soluble carbohydrates (WSC) concentration was lower (treatment $\times$ ensiling duration [TRT $\times$ D]; $P < 0.01$) for INO-L right after inoculation (d 0), but no treatment effect was observed after 30 and 90 d of ensiling. Yeast count was lower for INO-L after 90 d of ensiling (TRT $\times$ D; $P < 0.01$); however, it did not differ from CON-L at d 0 and after 30 d of ensiling. No treatment effect was observed for mold count (Table 1).

No treatment effect was observed for silage pH regardless of day. Additionally, no treatment effect was observed for lactate concentration ($P = 0.63$); however, both acetate (TRT $\times$ D; $P < 0.01$) and 1,2-propanediol concentrations were greater with inoculation ($P < 0.01$), and the magnitude of increase was greater after 90 d of ensiling (TRT $\times$ D; $P < 0.01$). Ammonia-N concentrations were greater (0.81% vs. 0.69% of total N; $P < 0.01$), whereas DM loss was lower (1.06% vs. 3.74%; $P < 0.01$) for INO-L than for CON-L (Table 2). Aerobic stability increased after 90 d of ensiling for INO-L compared with CON-L (TRT $\times$ D; $P < 0.01$; Figure 1).

Experiment 2

The ingredients and chemical composition of the prepartum and postpartum TMR are presented in Table 3. Supplemental Tables S1 and S2 (see Notes) show the chemical composition, fermentation profile, and degradability of corn and ryegrass ensiled for 395 and 98 d, respectively, using Ag-Bags. We also evaluated the aerobic stability of the TMR in the pre- and postpartum phases (Supplemental Figure S1, see Notes).

Production Parameters, Intake, and Rumination Time

Feeding INO did not affect DMI ($P = 0.50$), ADG ($P = 0.50$), BW ($P = 0.31$), or rumination ($P = 0.53$) during the prepartum period. Similarly, during the postpartum phase, no treatment effects were observed on MY ($P = 0.16$), or feed efficiency, regardless of adjusting for ECM, FCM, or MY. However, FCM yield increased by 3.5% (P

Table 1. Effect of microbial inoculation and ensiling duration on the nutritive value of corn silage (experiment 1)

Item	Treatment ¹						SEM	P-value		
	CON-L			INO-L				TRT	D	T × D ²
	0 d	30 d	90 d	0d	30 d	90 d				
DM, % of fresh matter	40.8	38.8	38.7	40.3	38.9	39.2	0.56	0.85	<0.01	0.41
OM, % DM	96.7	97.1	97.0	96.6	97.1	97.0	0.09	0.89	<0.01	0.78
aNDF, ³ % DM	42.4	42.6	39.1	43.8	45.3	38.3	1.08	0.23	<0.01	0.29
ADF, % DM	19.4	20.0	18.6	20.1	19.8	19.1	0.61	0.54	0.21	0.69
CP, ⁴ % DM	—	7.44	7.41	—	7.47	7.40	0.04	0.82	0.23	0.64
Starch, % DM	33.4	33.8	35.9	30.1	33.2	36.7	1.16	0.20	<0.01	0.11
WSC, % DM	10.1 ^a	2.09 ^c	2.03 ^c	8.98 ^b	2.05 ^c	1.89 ^c	0.20	<0.01	<0.01	<0.01
Ash, % DM	3.33	2.88	2.99	3.35	2.90	2.96	0.09	0.89	<0.01	0.78
Yeast count, log cfu/g	3.71 ^{ab}	4.57 ^a	3.63 ^b	3.79 ^{ab}	4.54 ^a	2.07 ^c	0.34	0.06	<0.01	0.03
Mold count, log cfu/g	3.55	4.67	2.72	3.53	4.44	1.92	0.46	0.27	<0.01	0.58

^{a-c}Mean values in the same row with different superscripts differ according to the P -value for the interaction effect.

¹Fresh forage was harvested from different plots ($n = 5$) and harvested forage from each plot was split into two 50-kg piles, placed on separate plastic sheets, and treated with either double-distilled water (CON), or a mixture of *Lactococcus lactis* and *Lentilactobacillus buchneri* (SiloSolve FC, Chr Hansen) containing 150,000 cfu/g applied at 30 mL/kg of forage (INO) and ensiled for 30 and 90 d.

²T $\times$ D = effect of interaction between treatment and ensiling duration.

³aNDF = amylase-corrected NDF.

⁴Day 0 was not analyzed for CP.

Table 2. Effect of microbial inoculation and ensiling duration on the fermentation characteristics of corn silage (experiment 1)

Item	Treatment ¹					<i>P</i> -value		
	CON-L		INO-L		SEM			
	30 d	90 d	30 d	90 d		TRT	D	T × D ²
pH	3.77	3.80	3.75	3.78	0.02	0.16	0.05	0.75
Lactate, % DM	5.27	5.77	5.21	5.69	0.15	0.63	<0.01	0.94
Acetate, % DM	1.00 ^c	1.17 ^b	1.18 ^b	1.72 ^a	0.05	<0.01	<0.01	<0.01
Ethanol, % DM	0.89	0.89	0.90	0.78	0.10	0.37	0.30	0.29
1,2-Propanediol, % DM	0.01 ^c	0.02 ^c	0.10 ^b	0.47 ^a	0.02	<0.01	<0.01	<0.01
Ammonia-N, % of total N	0.57	0.81	0.69	0.93	0.03	<0.01	<0.01	0.97
DM loss, %	3.80	3.68	0.86	1.16	0.61	<0.01	0.87	0.70

^{a-c}Mean values in the same row with different superscripts differ according to the *P*-value for the interaction effect.

¹Fresh forage was harvested from different plots (*n* = 5) and harvested forage from each plot was split into two 50-kg piles, placed on separate plastic sheets, and treated with either double-distilled water (CON), or a mixture of *Lactococcus lactis* and *Lentilactobacillus buchneri* (SiloSolve FC, Chr Hansen) containing 150,000 cfu/g applied at 30 mL/kg of forage (INO) and ensiled for 30 and 90 d.

²T × D = effect of interaction between treatment and ensiling duration.

= 0.03), ECM yield by 0.5% (*P* = 0.05), and milk fat yield was increased by 8% for cows fed INO. Additionally, we observed a tendency for treatment-by-week interaction for DMI, as feeding INO silage-based TMR tended to increase DMI at ~5 wk after calving (*P* = 0.07, Figure 2). Milk fat yield was greater (*P* = 0.03) for cows fed INO compared with cows fed CON, and a treatment-by-week interaction for fat concentration was observed (*P* < 0.01, Figure 3). Milk lactose (*P* = 0.91) and milk true protein concentrations (*P* = 0.64), along with yields of milk true protein (*P* = 0.28) and milk lactose (*P* = 0.19), were not different between treatments. Cows fed INO tended to have increased rumination time (536 vs. 511 min, *P* = 0.08). No treatment effects were observed on MUN (*P* = 0.84) and SCS (*P* = 0.39). Similarly, calf BW was similar between both treatments (*P* = 0.99; Table 4).

Apparent Nutrient Digestibility and Rumen Fermentation

No treatment effects were observed on DM (*P* = 0.73), CP (*P* = 0.11), NDF (*P* = 0.48), or ADF (*P* = 0.45) digestibility during the prepartum period (Table 5). Similarly, for the postpartum period, DM (*P* = 0.38), CP (*P* = 0.75), and ADF (*P* = 0.55) digestibility were not affected by inoculation. However, NDF (*P* = 0.06) digestibility tended to increase when cows were fed INO (Table 5).

Prepartum cows fed INO had greater concentrations of ruminal NH₃-N (*P* < 0.01; Table 6); however, no effects were observed on ruminal pH (*P* = 0.94); proportions of acetate (*P* = 0.74), propionate (*P* = 0.15), butyrate (*P* = 0.56), isobutyrate (*P* = 0.56), valerate (*P* = 0.42), or isovalerate (*P* = 0.47); or the acetate-to-propionate ratio (*P* = 0.16). During the postpartum period, no treatment effect was observed on NH₃-N (*P* = 0.32), rumen pH (*P* = 0.22),

or total VFA concentrations (*P* = 0.60). Similarly, proportions of ruminal acetate (*P* = 0.76), propionate (*P* = 0.13), butyrate (*P* = 0.38), isobutyrate (*P* = 0.34), and isovalerate (*P* = 0.41) did not differ between treatments. However, valerate (*P* = 0.06) proportions tended to decrease for cows fed INO. No treatment effects were observed on the acetate-to-propionate ratio (*P* = 0.33; Table 6).

Urinary Metabolites and Nitrogen Metabolism

Based on data collected from 7 blocks in this study, feeding INO had no effects on allantoin (*P* = 0.75), uric acid (*P* = 0.15), PD concentration (*P* = 0.67), excreted (*P* = 0.74) and absorbed purines (*P* = 0.74), creatinine concentration (*P* = 0.33), or total urinary output (*P* = 0.41) in prepartum cows. Similarly, for postpartum cows, no treatment effects were observed on allantoin (*P* = 0.84), uric acid (*P* = 0.78), PD concentration (*P* = 0.80), or excreted (*P* = 0.25) and absorbed purines (*P* = 0.24). However, creatinine concentration (*P* = 0.07) tended to decrease, whereas total urine output (*P* = 0.07) tended to increase in cows fed INO (Table 7).

No treatment differences were observed in N intake (*P* = 0.13), milk N (*P* = 0.37), milk N as a percentage of N intake (*P* = 0.21), microbial N (*P* = 0.58), fecal N output (*P* = 0.78), or urine N (*P* = 0.95). Greater N retention was observed in lactating dairy cows fed INO (*P* = 0.04; Table 8).

Plasma Metabolites

Blood parameters, including NEFA (*P* = 0.20), glucose (*P* = 0.82), and PUN (*P* = 0.53), were not different between treatments for prepartum cows. Similarly, no treatment effects were observed on NEFA (*P* = 0.77),

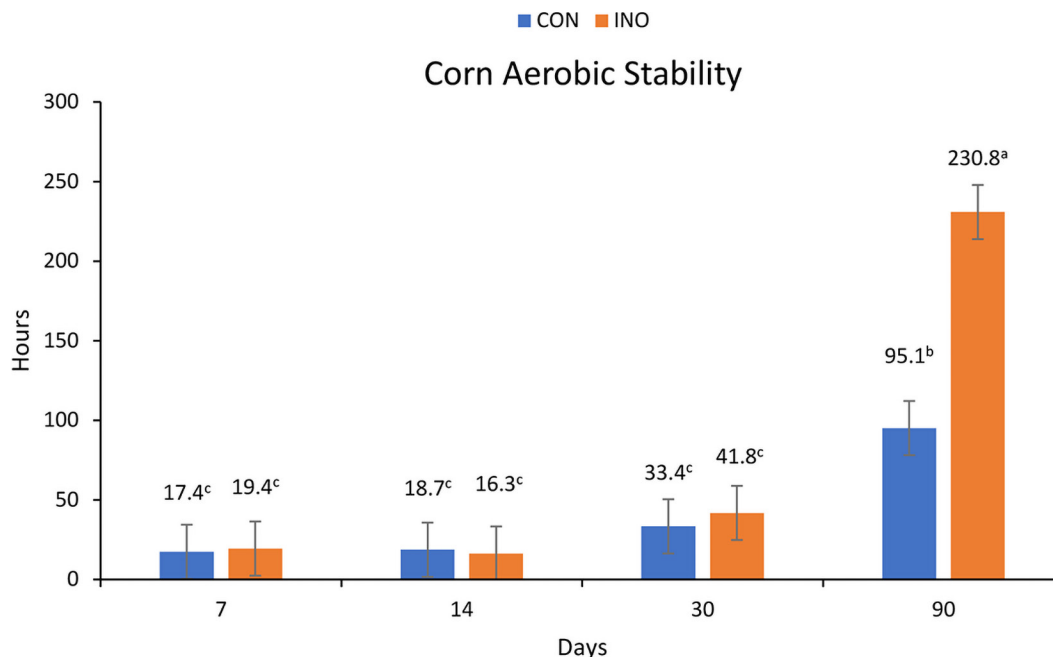


Figure 1. The effect of microbial inoculation on aerobic stability of corn silage preserved for 7, 14, 30, and 90 d under laboratory conditions (experiment 1; treatment: $P < 0.01$; day: $P < 0.01$; treatment $\times$ day: $P < 0.01$). Aerobic stability was determined when the difference between the silage and ambient temperatures was greater than 2°C. Error bars represent SEM. Mean values with different letters (a–c) differ according to the P -value for the interaction effect.

glucose ($P = 0.12$), and PUN concentrations ($P = 0.46$) for postpartum cows (Table 9).

DISCUSSION

Effect of Microbial Inoculation on Silage Fermentation

The effects of microbial inoculation on fermentation characteristics, nutritive value, and aerobic stability have

been well-researched (Reich and Kung, 2010; Oliveira et al., 2017; Nair et al., 2020). Facultative heterofermentative and obligate homofermentative lactic acid bacteria (LAB) ferment hexoses such as glucose into lactic acid. Sucu and Filya (2006) reported a lower concentration of WSC with LAB-inoculated corn after 2 d of ensiling; however, Gallo et al. (2018) reported no differences between inoculated and uninoculated corn before ensiling. We still lack enough data on the immediate effects (d 0) of microbial inoculation of corn silage on fermentation

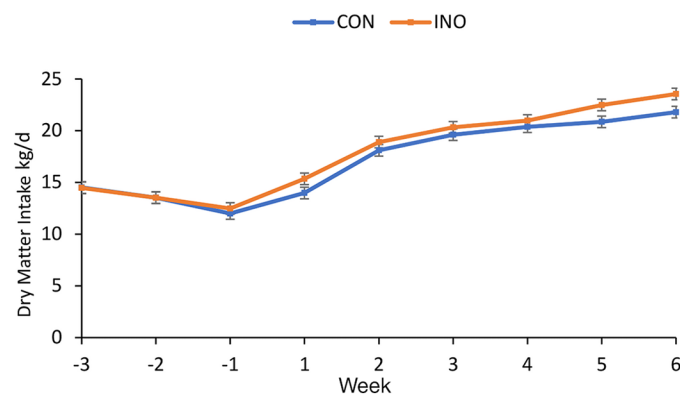


Figure 2. The effects of feeding inoculated silage-based TMR on DMI in transition dairy cows from 3 wk prepartum (treatment: $P = 0.50$; week: $P = 0.13$; treatment $\times$ week: $P = 0.94$) to 6 wk postpartum (treatment: $P = 0.13$; week: $P < 0.01$; treatment $\times$ week: $P = 0.07$). Error bars represent SEM.

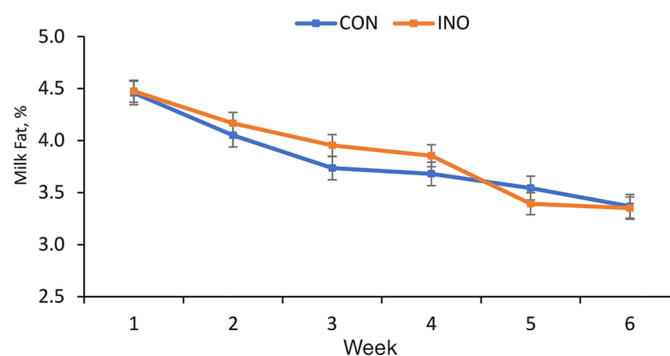


Figure 3. The effects of feeding inoculated silage-based TMR on milk fat concentration in lactating dairy cows (experiment 2; treatment: $P = 0.68$; week: $P < 0.01$; treatment $\times$ week: $P < 0.01$). Error bars represent SEM.

Table 3. Ingredient and chemical composition of the experimental diets (experiment 2)¹

Item	Prepartum		Postpartum	
	CON	INO	CON	INO
Ingredient, % of diet DM				
Corn silage	53.9	53.9	43.0	43.0
Ryegrass silage	9.29	9.29	9.68	9.68
Grass hay	8.57	8.57	—	—
Corn grain, ground shelled	—	—	12.9	12.9
Soybean meal, 44%	12.9	12.9	14.5	14.5
Soyhulls	7.14	7.14	6.82	6.82
Amino plus ²	—	—	4.30	4.30
Nurisol ³	—	—	0.64	0.64
Mineral and vitamin mix (prepartum) ⁴	8.21	8.21	—	—
Mineral and vitamin mix (postpartum) ⁵	—	—	6.00	6.00
Molasses	—	—	2.15	2.15
Nutrient composition, % of DM				
NEL, ⁶ Mcal/kg	1.46	1.46	1.36	1.36
OM, %	91.3	91.6	92.6	92.5
CP, %	14.2	14.4	17.3	17.1
aNDF, %	32.0	33.0	27.6	28.9
Forage aNDF, %	23.0	23.7	15.8	16.5
ADF, %	18.4	19.0	15.8	16.8
NFC, ⁷ %	45.1	44.6	49.6	45.3
Crude fat, %	2.38	2.34	2.45	2.98
Ash, %	8.69	8.43	7.39	7.50

¹Diets pre- and postpartum contained corn and ryegrass silage either inoculated with water (CON) or *L. lactis* and *L. buchneri* with 150,000 cfu/g (INO).

²Trademarked soybean meal (Ag Processing Inc., Omaha, NE).

³Nurisol (calcium salt of long-chain fatty acids), Global Agri-Trade Organization (Rancho Dominguez, CA).

⁴Vitamin mineral mixture (DM basis; prepartum diets): Ca, 16.5%; P, 0.26%; Mg, 3.90%; K, 0.32%, S, 2.78%; Na, 1.17%; Cl, 6.88%; Fe 175.8 mg/kg; Zn 823.05 mg/kg; Cu, 127.13 mg/kg; Mn, 764.07 mg/kg; Se, 4.16 mg/kg; Co, 12.35 mg/kg; I, 7.06 mg/kg; vitamin A, 131.67 kIU/kg; vitamin D, 16.41 kIU/kg; vitamin E, 302.4 IU/kg; biotin, 1.23 mg/kg; choline, 17,766 mg/kg.

⁵Vitamin mineral mixture (DM basis; postpartum diets): Ca, 7.32%; P, 1.62%; Mg, 2.49%; K, 0.22%, S, 0.42%; Na, 8.19%; Cl, 3.33%; Fe 1,236.6 mg/kg; Zn 1,213.4 mg/kg; Cu, 184.2 mg/kg; Mn, 995.1 mg/kg; Se, 6.15 mg/kg; Co, 18.89 mg/kg; I, 10.37 mg/kg; vitamin A, 100.9 kIU/kg; vitamin D, 24.16 kIU/kg; vitamin E, 445.2 IU/kg; biotin, 1.81 mg/kg.

⁶Estimated value (NRC, 2001).

⁷NFC estimated based on 100 – (CP + NDF + fat + ash).

parameters; however, with the lower concentration of WSC observed in the current study, we can speculate that microbial inoculants, including LAB, have immediate effects.

The primary purpose of using inoculants in the ensiling process is to facilitate and increase lactic acid production, thereby rapidly decreasing silage pH and inhibiting the proliferation of detrimental microorganisms. Lactic acid bacteria use WSC as substrates to produce lactic acid, explaining the effect of lower WSC, greater lactate concentrations, and lower silage pH for INO-L. Similar results were obtained in a previous study assessing the effect of dual-purpose inoculants in farm-scale silos (Queiroz et al., 2012).

Heterolactic bacteria utilize lactic acid produced by the homolactic bacteria during fermentation and produce acetate and 1,2-propanediol (Oude Elferink et al., 2001), which are known for their antifungal activity (Muck et al., 2018). In agreement with previous studies (Gallo et

al., 2018, 2021; Saylor et al., 2020), we observed greater acetate and 1,2-propanediol with the inoculation of *L. buchneri* and *L. lactis*. Greater aerobic stability observed with inoculated silage may be attributed to increased concentrations of acetic acid and its antifungal properties (Muck et al., 2018). Aerobic stability measures the time the silage remains stable after aerobic exposure. Spoilage during the feed-out phase occurs primarily due to yeast-mediated degradation of lactic acid, thereby increasing the pH and allowing the growth of deleterious bacteria and molds, lowering silage quality (Kung, 2010; Gallo et al., 2018).

Microbial inoculants using combined strains have reduced DM losses, probably by improving fermentation efficiency compared with uninoculated silage (Drouin et al., 2019), and similar findings from our study further support this speculation. The primary product of homo-fermentative bacteria is lactic acid, and because CO₂ is not produced during this process, DM losses are lower

Table 4. Effect of feeding silage-based TMR containing corn silage and ryegrass silage ensiled with (INO) or without (CON) microbial inoculant on the production parameters of lactating dairy cows (experiment 2)

Item	Treatment ¹		SEM	TRT	P-value	
	CON	INO			Week	T × W ²
Prepartum						
DMI, kg/d	13.3	13.8	0.58	0.50	0.13	0.94
ADG, kg/d	1.27	0.85	0.35	0.50	—	—
BW, kg	805	788	10.2	0.31	0.14	0.27
Rumination, min/d	495	515	11.1	0.53	0.05	0.37
Postpartum						
DMI, kg/d	19.1	20.3	0.54	0.13	<0.01	0.07
Yield, kg/d						
Milk	41.0	43.2	1.27	0.16	<0.01	0.72
3.5% FCM ³	42.0	44.9	1.14	0.03	<0.01	0.57
ECM ⁴	41.9	44.5	1.15	0.05	<0.01	0.74
Milk fat	1.50	1.62	0.05	0.03	<0.01	0.12
Milk true protein	1.25	1.31	0.04	0.28	<0.01	0.89
Milk lactose	1.98	2.09	0.07	0.19	<0.01	0.79
Concentration, %						
Milk fat	3.81	3.87	0.11	0.68	<0.01	<0.01
Milk true protein	3.14	3.12	0.05	0.64	<0.01	0.02
Milk lactose	4.81	4.81	0.04	0.91	<0.01	<0.01
Efficiency						
Milk yield/DMI	2.19	2.17	0.06	0.82	<0.01	0.69
3.5% FCM/DMI	2.18	2.20	0.07	0.82	<0.01	0.13
ECM/DMI	2.26	2.26	0.07	0.88	<0.01	0.24
MUN, mg/dL	8.67	8.84	0.53	0.84	<0.01	<0.01
SCS ⁵	4.43	4.00	0.36	0.39	<0.01	<0.01
Rumination, min/d	511	536	12.1	0.08	<0.01	0.18
Calf BW, kg	41.6	40.9	5.5	0.99	—	—

¹Cows were fed either CON (a TMR composed of uninoculated corn and ryegrass silage) or INO (a TMR composed of inoculated corn and ryegrass silage).

²T × W = effect of interaction between treatment and week of study.

³FCM = 0.4324 × milk yield (kg/d) + 16.216 × fat yield (kg/d).

⁴ECM = 0.327 × milk yield (kg/d) + 12.95 × fat yield (kg/d) + 7.65 × true protein yield (kg/d).

⁵SCS = log₁₀(SCC/12.5)/log₁₀ of 2.

Table 5. Effects of feeding silage-based TMR containing corn silage and ryegrass silage ensiled with (INO) or without (CON) microbial inoculant on the total-tract apparent nutrient digestibility of transition dairy cows (experiment 2)

Nutrient digestibility, % ²	Treatment ¹		SEM	P-value		
	CON	INO		Treatment	Week	T × W ³
Prepartum						
DM	62.0	61.6	0.71	0.73	0.01	0.04
CP	58.3	54.1	1.88	0.11	0.24	0.50
NDF	43.5	42.7	0.77	0.48	0.00	0.06
ADF	39.5	40.7	1.21	0.45	0.01	0.21
Postpartum						
DM	68.2	67.1	0.92	0.38	0.07	0.69
CP	57.4	56.5	2.15	0.75	0.66	0.83
NDF	42.0	45.2	1.23	0.06	0.65	0.45
ADF	40.5	41.6	1.37	0.55	0.51	0.86

¹Cows were fed either CON (a TMR composed of uninoculated corn and ryegrass silage) or INO (a TMR composed of inoculated corn and ryegrass silage).

²Samples for digestibility estimations were collected during wk −3 and −1 relative to the expected calving date, and wk 2 and 6 postpartum.

³T × W = effect of interaction between treatment and week of study.

Table 6. Effects of feeding silage-based TMR containing corn silage and ryegrass silage ensiled with (INO) or without (CON) microbial inoculant on the ruminal fermentation of transition dairy cows (experiment 2)¹

Item	Treatment ²		SEM	P-value		
	CON	INO		TRT	Week	T × W ³
Prepartum						
pH	6.63	6.63	0.07	0.94	0.03	0.48
Ammonia-N, mM	4.18 ^b	5.47 ^a	0.34	<0.01	0.55	0.34
Total VFA, mM	114	115	4.16	0.81	0.79	0.99
Individual VFA, mol/100 mol						
Acetate	54.8	58.3	1.02	0.74	0.97	0.79
Propionate	22.2	21.1	0.52	0.15	0.23	0.23
Butyrate	12.9	12.5	0.49	0.56	0.52	0.05
Isobutyrate	2.83	3.22	0.47	0.56	0.81	0.91
Valerate	1.51	1.71	0.17	0.42	0.80	0.58
Isovalerate	2.66	2.93	0.26	0.47	0.73	0.20
Acetate:propionate	2.68	2.87	0.09	0.16	0.30	0.39
Postpartum						
pH	6.02	6.18	0.09	0.22	0.93	0.86
Ammonia-N, mM	3.59	4.34	0.57	0.32	0.15	0.76
Total VFA, mM	123	118	6.74	0.6	0.99	0.91
Individual VFA, mol/100 mol						
Acetate	55.1	56.0	2.26	0.76	0.03	0.23
Propionate	24.9	28.1	1.57	0.13	0.12	0.16
Butyrate	12.6	10.7	1.58	0.38	0.21	0.87
Isobutyrate	2.26	1.76	0.40	0.34	0.44	0.27
Valerate	2.51	1.47	0.40	0.06	0.49	0.68
Isovalerate	2.42	1.96	0.42	0.41	0.50	0.56
Acetate:propionate	1.91	2.05	0.12	0.33	0.32	0.89

^{a,b}Mean values in the same row with different superscripts differ according to the *P*-value for the treatment effect.

¹Rumen fluid samples were collected using a stomach tube every week during the prepartum period (−3, −2, and −1 wk) and during wk 2 and 6 in the postpartum period, approximately 2 h after feeding.

²Cows were fed either CON (a TMR composed of uninoculated corn and ryegrass silage) or INO (a TMR composed of inoculated corn and ryegrass silage).

³T × W = effect of interaction between treatment and week of study.

Table 7. Effect of feeding silage-based TMR containing corn silage and ryegrass silage ensiled with (INO) or without (CON) microbial inoculant on urinary metabolites of transition dairy cows (experiment 2)

Item	Treatment ¹		SEM	P-value		
	CON	INO		TRT	Week	T × W ²
Prepartum						
Allantoin, mmol/L	9.43	9.71	0.60	0.75	0.19	0.63
Uric acid, mmol/L	0.87	0.99	0.06	0.15	0.29	0.44
PD concentration, mmol/L	10.4	10.7	0.62	0.67	0.23	0.56
PD excretion, mmol/d	202	210	13.7	0.74	0.66	0.99
Absorbed purines, mmol/d	146	158	19.3	0.74	0.73	0.99
Creatinine, mmol/L	8.42	8.95	0.44	0.33	0.05	0.60
Urine output, ³ L/d	21.2	19.6	1.47	0.41	0.18	0.79
Postpartum						
Allantoin, mmol/L	12.1	11.8	0.87	0.84	0.95	0.13
Uric acid, mmol/L	1.27	1.33	0.12	0.78	0.03	0.97
PD concentration, mmol/L	13.2	14.0	1.63	0.80	0.78	0.32
PD excretion, mmol/d	318	356	24.4	0.25	0.21	0.46
Absorbed purines, mmol/d	292	337	28.4	0.24	0.19	0.46
Creatinine, mmol/L	6.27	5.54	0.35	0.07	0.02	0.30
Urine output, ³ L/d	24.6	27.5	1.16	0.07	0.03	0.36

¹Cows were fed either CON (a TMR composed of uninoculated corn and ryegrass silage) or INO (a TMR composed of inoculated corn and ryegrass silage).

²T × W = effect of interaction between treatment and week of study.

³Spot urine samples were collected twice daily at 0600 and 1400 h for 2 consecutive days every week (−3, −2, and −1 wk) in the prepartum period and during wk 2 and 6 in the postpartum period.

Table 8. Effect of feeding silage-based TMR containing corn silage and ryegrass silage ensiled with (INO) or without (CON) microbial inoculant on the nitrogen metabolism in lactating dairy cows (experiment 2)

Item	Treatment ¹		SEM	P-value		
	CON	INO		TRT	Week	T × W ²
N Intake, g/d	422	519	36.0	0.13	0.06	0.74
Milk N, g/d	185	204	12.3	0.37	0.96	0.12
Milk N, % of N intake	46.4	42.0	2.58	0.21	0.03	0.21
Microbial N, g/d	211	237	27.6	0.58	0.11	0.05
Fecal N output g/d	227	210	30.2	0.78	0.01	0.80
Urine N, g/d	176	199	16.4	0.33	0.27	0.89
N retained g/d	-138	-51.0	27.3	0.04	0.94	0.41

¹Cows were fed either CON (a TMR composed of uninoculated corn and ryegrass silage) or INO (TMR composed of inoculated corn and ryegrass silage) Only 7 blocks were used for the analysis of N metabolism data.

²T × W = effect of interaction between treatment and week of study.

when homolactic inoculants are used. On the contrary, heterofermentative bacterial inoculants produce 1 mol of CO₂ and 1 mol of acetic acid, resulting in greater aerobic stability with increased DM losses. Using a combination of homo- and heterofermentative bacterial inoculants seems to enhance synergy in improving fermentation and preserving DM by inducing a rapid decline in silage pH and increasing acetic acid concentration to improve the aerobic stability later in the feed-out phase.

We found greater concentrations of NH₃-N for INO-L silage, suggesting increased degradation of the starch-protein matrix. Plant and microbial proteases can degrade plant proteins to peptides and amino acids, and silage microbes subsequently degrade amino acids to NH₃-N (Grum et al., 1991; Muck et al., 2003). Hence, the accumulation of NH₃-N is indicative of proteolysis as zein proteins are degraded with plant and microbial proteases (Hoffman et al., 2011; Saylor et al., 2020).

A known limitation of studies that measure fermentation parameters in farm-scale silos is the number of experimental units. Oftentimes, it is difficult to have an adequate number of experimental units due to the infeasibility of having multiple Ag-Bag silos for different

treatments. In this study, we did not statistically analyze the fermentation parameters data because we had only 2 experimental units per treatment, which aligns with the limitations stated previously. However, we present supplemental tables (see Notes) with chemical composition to help visualize our outcomes from inoculating farm-scale silos.

Effect of Feeding Inoculated Silage on Transition Dairy Cattle Performance

Studies evaluating the effects of feeding inoculated silage on the performance of lactating dairy cows are underrepresented, and to the best of our knowledge, this is the first study evaluating the effects of supplementing a combination of inoculated corn and ryegrass silage on the performance of transition dairy cows. However, these findings should be interpreted with caution because only a single corn hybrid and a single ryegrass variety were evaluated. We hypothesized that cows fed INO would increase DMI and, subsequently, the performance of lactating dairy cows due to the preserved quality and aerobic stability of the TMR. Oliveira et al. (2017)

Table 9. Effects of feeding silage-based TMR containing corn silage and ryegrass silage ensiled with (INO) or without (CON) microbial inoculant on the plasma metabolites of transition dairy cows (experiment 2)

Item	Treatment ¹		SEM	P-value		
	CON	INO		TRT	Week	T × W ²
Prepartum						
NEFA, mM	198	152	24.8	0.20	0.14	0.96
Glucose, mM	3.40	3.40	0.10	0.82	0.19	0.65
BUN, mg/dL	5.40	5.50	0.10	0.53	0.97	0.52
Postpartum						
NEFA, mM	621	594	66.8	0.77	<0.01	0.35
Glucose, mM	3.06	3.43	0.17	0.12	<0.01	0.21
BUN, mg/dL	5.27	5.36	0.09	0.46	0.29	0.66

¹Cows were fed either CON (a TMR composed of uninoculated corn and ryegrass silage) or INO (a TMR composed of inoculated corn and ryegrass silage).

²T × W = effect of interaction between treatment and week of study.

reported that dairy cows consuming silage inoculated with homofermentative or facultative heterofermentative bacteria exhibited greater performance. Similarly, Daniel et al. (2018) observed increased MY and FCM yield with feeding corn silage inoculated with *L. lactis*, *Lactobacillus plantarum*, and *Enterococcus faecium*. In our study, we found greater 3.5% FCM and ECM yields for cows fed INO. These increases may be attributed to improved DMI in wk 5 and 6 of the experimental period for cows fed INO, as indicated by a tendency in the interaction effect when DMI was analyzed by week. Previous studies (Blajman et al., 2018; Saylor et al., 2020; Arriola et al., 2021) reported increased acetate levels in silages preserved with *L. buchneri*. Feeding silage with a greater proportion of acetic acid, a known precursor for milk fat synthesis, is effective at enhancing its net portal flux and milk fat synthesis (Aschenbach et al., 2011). In our study, cows fed INO showed greater overall fat yield. Although a treatment-by-week interaction was reported for fat concentration, no significant differences were detected when analyzed within individual weeks. The limitation of being unable to statistically analyze fermentation parameters, as stated previously, prevents us from establishing acetic acid as a valid explanation for the increased fat yield. Alternatively, this difference may be related to increased DMI observed in the last 2 wk of the experimental period.

Rumination time was increased for lactating cows fed INO, and it may have influenced NDF digestibility and, subsequently, milk fat output. Rumination is defined as the regurgitation and remastication of ruminal digesta occurring between meals and during resting to break large feed particles and increase microbial attachment and digestion (Beauchemin, 2018). It is well-established that changes in dietary composition influence rumination and chewing behavior (Beauchemin, 2018). Previous studies have observed increased rumination time with greater dietary NDF (Beauchemin and Buchanan-Smith, 1989) or with increasing physically effective NDF concentration (White et al., 2017). This relationship was further confirmed in a recent meta-analytical study (Souza et al., 2022) that observed a quadratic relationship between rumination time and total-tract NDF digestibility (TTNDFd) and milk fat yield as rumination time increased with TTNDFd. Maximum digestibility (52.3%) was predicted at 443 min/d rumination time, whereas a little gain was observed in milk fat yield beyond a rumination time of ~450 min/d. Oba and Allen (1999) evaluated the relationship between changes in NDF digestibility and animal responses. They reported that a 1-unit increase in NDF digestibility was positively associated with a 0.25-kg increase of 3.5% FCM yield. Based on observations from previous studies, we can conclude that greater rumination observed in cows fed INO in the cur-

rent study may have improved rumen pH, enhanced fiber digestion, and increased milk fat yield.

Daniel et al. (2018) observed greater DM digestibility and a tendency for greater TTNDFd with inoculated silage. Similarly, Adesogan et al. (2009) observed increased DM digestibility with silages inoculated with homofermentative or facultative heterofermentative inoculants. Similar improvements in DM digestibility have been observed earlier using in vitro (Weinberg et al., 2007; Cao et al., 2011) and in vivo (Kung and Muck, 1997; Ando et al., 2006) experimental models in response to microbial inoculation during ensiling. However, a specific mechanism resulting in this improvement has not yet been identified. Perhaps the hydrolytic effect of enzymes produced by LAB may increase diet digestibility. Daniel et al. (2018) found that feeding homofermentative bacterial-inoculated corn silage to lactating cows resulted in a ~4% increase in TTNDFd, which is very similar to the ~3.2% increase in TTNDFd observed in the present study. It is possible that the tendency for greater NDF digestibility for cows fed INO may have increased the net portal flux of nutrients to support the increased FCM and ECM observed in this study.

Urine and Nitrogen Metabolism

We found no differences in urinary metabolites between treatments during the prepartum period. However, in the postpartum period, we found that the urine output tended to increase for cows fed INO. Dietary CP concentration has been shown to affect urine volume (Broderick, 2003; Wattiaux and Karg, 2004), as increased urine volume is necessary for removing excess N consumed by dairy cows (Holter et al., 1982). Although we found no significant difference in N intake ($P = 0.13$), cows fed INO consumed 97 g/d more N than cows fed CON, which may explain the increase in urine output. This higher N intake for the cows fed INO is supported by the increase in DMI (1.2 kg/d). Inoculation of silages promotes protein conservation, which could be translated into an increased ruminal microbial protein synthesis (Oliveira et al., 2017; Kok et al., 2024). Therefore, N output in milk could also increase when inoculated silages are fed, explaining our findings, which although not significant, show that milk N increased (~20g/d) for cows fed INO. It is important to note that the N metabolism evaluations were based on only 7 blocks; therefore, results involving N intake, excretion, and related inferences should be interpreted with caution.

Although we reported a significant interaction, there was no effect of treatment on microbial N for wk 2 and 6 of the experimental period. Retention of N was negative for both treatments, meaning that cows excreted more N than they consumed. However, cows fed INO had a less

negative N balance ($P = 0.04$). Similarly, due to the improved protein preservation caused by inoculation, cows fed INO likely had greater ruminal ammonia availability and nitrogen recycling (Kok et al., 2024), which could indicate a potential for greater capacity to improve N utilization efficiency. Furthermore, this difference may indicate that the cows fed INO had a lower negative nutrient balance. The plasma metabolite results confirm this lower degree of negative nutrient balance. We found that although the difference was not significant, cows fed INO had lower NEFA in the pre- and postpartum periods. It is well known that cows with a negative energy balance mobilize adipose tissue reserves to cope with the energy required for milk production during the transition period. When this fat mobilization is in excess, metabolites of fat degradation, such as NEFA, increase in the blood (Grummer, 2008).

CONCLUSIONS

Microbial inoculation of corn and ryegrass silage is a promising approach to improving the fermentation characteristics, nutritive value, and aerobic stability. These improvements, in turn, can potentially enhance the performance of lactating dairy cows by supporting digestibility and milk production. Future research should investigate the mechanisms underlying these effects and explore the potential for optimizing inoculant formulations to maximize silage production and benefits for dairy cattle.

NOTES

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Nonstandard abbreviations used: aNDF = amylase-treated NDF; AP = absorbed purine; CON = TMR composed of uninoculated corn and ryegrass silage; CON-AC = uninoculated corn silage; CON-AR = uninoculated ryegrass silage; CON-L = corn forage inoculated with

double-distilled water; INO = TMR composed of inoculated corn and ryegrass silage; INO-AC = inoculated corn silage; INO-AR = inoculated ryegrass silage; INO-L = corn forage inoculated with mixed microbial inoculant; LAB = lactic acid bacteria; MY = milk yield; NEFA = nonesterified fatty acid; PD = purine derivative; PUN = plasma urea nitrogen; RCBD = randomized complete block design; $TRT \times D$ = treatment $\times$ ensiling duration; TTNDFd = total-tract NDF digestibility; uNDF = undigested NDF.

REFERENCES

- Adesogan, A. T., and K. G. Arriola. 2020. Bacterial Inoculants for Optimizing Silage Quality and Preservation and Enhancing Animal Performance. Pages 42–57 in Proc. Cornell Nutr. Conf., Syracuse, NY. Cornell University, Ithaca, NY.
- Adesogan, A. T., L. E. Sollenberger, J. E. Moore, Y. C. Newman, and J. M. B. Vendramini. 2009. Factors affecting forage quality. University of Florida IFAS Extension. <https://doi.org/10.32473/edis-ag161-2009>.
- Ando, S., M. Ishida, S. Oshio, and O. Tanaka. 2006. Effects of isolated and commercial lactic acid bacteria on the silage quality, digestibility, voluntary intake and ruminal fluid characteristics. Asian-Australas. J. Anim. Sci. 19:386–389. <https://doi.org/10.5713/ajas.2006.386>.
- AOAC International. 1995. Official Methods of Analysis. 16th ed. AOAC International.
- AOAC International. 2000. Official Methods of Analysis of AOAC International. 17th ed. AOAC International.
- AOAC International. 2012. Official Methods of Analysis. 19th ed. AOAC International.
- Arriola, K. G. 2010. Using biological additives to improve dietary nutrient conservation and utilization by lactating dairy cows. PhD thesis. Department of Animal Sciences. Institute of Food and Agricultural Sciences. University of Florida, Gainesville, FL.
- Arriola, K. G., D. Vyas, D. Kim, M. C. N. Agarussi, V. P. Silva, M. Flores, Y. Jiang, X. Yanlin, A. A. Pech-Cervantes, L. F. Ferraretto, and A. T. Adesogan. 2021. Effect of *Lactobacillus hilgardii*, *Lactobacillus buchneri*, or their combination on the fermentation and nutritive value of sorghum silage and corn silage. J. Dairy Sci. 104:9664–9675. <https://doi.org/10.3168/jds.2020-19512>.
- Aschenbach, J. R., G. B. Penner, F. Stumpff, and G. Gäbel. 2011. Ruminant nutrition symposium: Role of fermentation acid absorption in the regulation of ruminal pH. J. Anim. Sci. 89:1092–1107. <https://doi.org/10.2527/jas.2010-3301>.
- Beauchemin, K. A. 2018. Invited review: Current perspectives on eating and rumination activity in dairy cows. J. Dairy Sci. 101:4762–4784. <https://doi.org/10.3168/jds.2017-13706>.
- Beauchemin, K. A., and J. G. Buchanan-Smith. 1989. Effects of dietary neutral detergent fiber concentration and supplementary long hay on chewing activities and milk production of dairy cows. J. Dairy Sci. 72:2288–2300. [https://doi.org/10.3168/jds.S0022-0302\(89\)79360-7](https://doi.org/10.3168/jds.S0022-0302(89)79360-7).
- Bernard, J. K., J. W. West, and D. S. Trammell. 2002. Effect of replacing corn silage with annual ryegrass silage on nutrient digestibility, intake, and milk yield for lactating dairy cows. J. Dairy Sci. 85:2277–2282. [https://doi.org/10.3168/jds.S0022-0302\(02\)74307-5](https://doi.org/10.3168/jds.S0022-0302(02)74307-5).
- Blajman, J. E., R. B. Páez, C. G. Vinderola, M. S. Lingua, and M. L. Signorini. 2018. A meta-analysis on the effectiveness of homofermentative and heterofermentative lactic acid bacteria for corn silage. J. Appl. Microbiol. 125:1655–1669. <https://doi.org/10.1111/jam.14084>.
- Broderick, G. A. 2003. Effects of varying dietary protein and energy levels on the production of lactating dairy cows. J. Dairy Sci. 86:1370–1381. [https://doi.org/10.3168/jds.S0022-0302\(03\)73721-7](https://doi.org/10.3168/jds.S0022-0302(03)73721-7).
- Canale, A., M. E. Valente, and A. Ciotti. 1984. Determination of volatile carboxylic acids (C_1 – C_{51}) and lactic acid in aqueous acid extracts

- of silage by high performance liquid chromatography. *J. Sci. Food Agric.* 35:1178–1182. <https://doi.org/10.1002/jsfa.2740351106>.
- Cao, Y., Y. Cai, T. Takahashi, N. Yoshida, M. Tohno, R. Uegaki, K. Nonaka, and F. Terada. 2011. Effect of lactic acid bacteria inoculant and beet pulp addition on fermentation characteristics and in vitro ruminal digestion of vegetable residue silage. *J. Dairy Sci.* 94:3902–3912. <https://doi.org/10.3168/jds.2010-3623>.
- Chen, X. B., and M. J. Gomes. 1992. Estimation of microbial protein supply to sheep and cattle based on urinary excretion of purine derivatives: An overview of the technical details. International Feed Resources Unit, Rowett Research Institute, Bucksburn Aberdeen, UK.
- Chizzotti, M. L., S. C. Valadares Filho, R. F. D. Valadares, F. H. M. Chizzotti, and L. O. Tedeschi. 2008. Determination of creatinine excretion and evaluation of spot urine sampling in Holstein cattle. *Livest. Sci.* 113:218–225. <https://doi.org/10.1016/j.livsci.2007.03.013>.
- Cobianchi, J. V., A. S. Oliveira, J. M. S. Campos, A. V. Guimarães, S. C. Valadares Filho, F. P. Cobianchi, and T. E. S. Oliveira. 2012. Productive performance and efficiency of utilization of the diet components in dairy cows fed castor meal treated with calcium oxide. *Rev. Bras. Zootec.* 41:2238–2248. <https://doi.org/10.1590/S1516-35982012001000015>.
- Daniel, J. L. P., O. C. M. Queiroz, K. G. Arriola, R. Daetz, F. Basso, J. J. Romero, and A. T. Adesogan. 2018. Effects of homolactic bacterial inoculant on the performance of lactating dairy cows. *J. Dairy Sci.* 101:5145–5152. <https://doi.org/10.3168/jds.2017-13880>.
- Drackley, J. K., T. R. Overton, and G. N. Douglas. 2001. Adaptations of glucose and long-chain fatty acid metabolism in liver of dairy cows during the periparturient period. *J. Dairy Sci.* 84:E100–E112. [https://doi.org/10.3168/jds.S0022-0302\(01\)70204-4](https://doi.org/10.3168/jds.S0022-0302(01)70204-4).
- Drouin, P., J. Tremblay, and F. Chaucheyras-Durand. 2019. Dynamic succession of microbiota during ensiling of whole plant corn following inoculation with *Lactobacillus buchneri* and *Lactobacillus hilgardii* alone or in combination. *Microorganisms* 7:595. <https://doi.org/10.3390/microorganisms7120595>.
- Ferraretto, L. F., R. D. Shaver, and B. D. Luck. 2018. Silage review: Recent advances and future technologies for whole-plant and fractionated corn silage harvesting. *J. Dairy Sci.* 101:3937–3951. <https://doi.org/10.3168/jds.2017-13728>.
- Gallo, A., T. F. Bernardes, G. Copani, P. Fortunati, G. Giuberti, S. Bruschi, K. A. Bryan, N. G. Nielsen, K. L. Witt, and F. Masoero. 2018. Effect of inoculation with *Lactobacillus buchneri* LB1819 and *Lactococcus lactis* O224 on fermentation and mycotoxin production in maize silage compacted at different densities. *Anim. Feed Sci. Technol.* 246:36–45. <https://doi.org/10.1016/j.anifeedsci.2018.09.009>.
- Gallo, A., F. Fancello, F. Ghilardelli, S. Zara, F. Foldi, and M. Spanghero. 2021. Effects of several lactic acid bacteria inoculants on fermentation and mycotoxins in corn silage. *Anim. Feed Sci. Technol.* 277:114962. <https://doi.org/10.1016/j.anifeedsci.2021.114962>.
- Grum, D. E., W. L. Shockey, and W. P. Weiss. 1991. Electrophoretic examination of alfalfa silage proteins. *J. Dairy Sci.* 74:146–154. [https://doi.org/10.3168/jds.S0022-0302\(91\)78155-1](https://doi.org/10.3168/jds.S0022-0302(91)78155-1).
- Grummer, R. R. 2008. Nutritional management strategies for the prevention of fatty liver in dairy cattle. *Vet. J.* 176:10–20. <https://doi.org/10.1016/j.tvjl.2007.12.033>.
- Hall, M. B. 2015. Determination of dietary starch in animal feeds and pet food by an enzymatic-colorimetric method: Collaborative study. *J. AOAC Int.* 98:397–409. <https://doi.org/10.5740/jaoacint.15-012>.
- Hoffman, P. C., N. M. Esser, R. D. Shaver, W. K. Coblenz, M. P. Scott, A. L. Bodnar, R. J. Schmidt, and R. C. Charley. 2011. Influence of ensiling time and inoculation on alteration of the starch-protein matrix in high-moisture corn. *J. Dairy Sci.* 94:2465–2474. <https://doi.org/10.3168/jds.2010-3562>.
- Hoffman, P. C., S. J. Sievert, R. D. Shaver, D. A. Welch, and D. K. Combs. 1993. In situ dry matter, protein, and fiber degradation of perennial forages. *J. Dairy Sci.* 76:2632–2643. [https://doi.org/10.3168/jds.S0022-0302\(93\)77599-2](https://doi.org/10.3168/jds.S0022-0302(93)77599-2).
- Holter, J. B., J. A. Byrne, and C. G. Schwab. 1982. Crude protein for high milk production. *J. Dairy Sci.* 65:1175–1188. [https://doi.org/10.3168/jds.S0022-0302\(82\)82329-1](https://doi.org/10.3168/jds.S0022-0302(82)82329-1).
- Huhtanen, P., K. Kaustell, and S. Jaakkola. 1994. The use of internal markers to predict total digestibility and duodenal flow of nutrients in cattle given six different diets. *Anim. Feed Sci. Technol.* 48:211–227. [https://doi.org/10.1016/0377-8401\(94\)90173-2](https://doi.org/10.1016/0377-8401(94)90173-2).
- Kok, I., G. Copani, K. A. Bryan, K. L. M. Witt, W. M. van Straalen, R. C. do Amaral, and B. I. Cappellozza. 2024. Effects of feeding an inoculated corn silage with or without a direct-fed microbial on dry matter intake, milk production, and nutrient digestibility of high-producing lactating Holstein cows. *Transl. Anim. Sci.* 8:txae010. <https://doi.org/10.1093/tas/txae010>.
- Kung, L., Jr. 2010. Aerobic stability of silage. In Proceedings of California Alfalfa and Forage Symposium and Crop/Cereal Conference, Visalia, CA. Accessed Feb. 7, 2026. https://alfalfasyposium.ucdavis.edu/symposium/2010/files/talks/CAS10_KungAnaerobicStability.pdf.
- Kung, L. J., and R. E. Muck. 1997. Effects of silage additives on ensiling. Pages 187–199 in *Silage: Field to Feedbunk*. Northeast Regional Agricultural Engineering Service (NRAES), Ithaca, NY.
- Muck, R. E. 1988. Factors influencing silage quality and their implications for management. *J. Dairy Sci.* 71:2992–3002. [https://doi.org/10.3168/jds.S0022-0302\(88\)79897-5](https://doi.org/10.3168/jds.S0022-0302(88)79897-5).
- Muck, R. E., L. E. Moser, and R. E. Pitt. 2003. Postharvest factors affecting ensiling. *Silage Sci. Tech.* 42:241–304. <https://doi.org/10.2134/agronmonogr42.c6>.
- Muck, R. E., E. M. G. Nadeau, T. A. McAllister, F. E. Contreras-Govea, M. C. Santos, and L. Kung Jr. 2018. Silage review: Recent advances and future uses of silage additives. *J. Dairy Sci.* 101:3980–4000. <https://doi.org/10.3168/jds.2017-13839>.
- Mulligan, F. J., and M. L. Doherty. 2008. Production diseases of the transition cow. *Vet. J.* 176:3–9. <https://doi.org/10.1016/j.tvjl.2007.12.018>.
- Nair, J., N. Huaxin, E. Andrada, H. E. Yang, E. Chevaux, P. Drouin, T. A. McAllister, and Y. Wang. 2020. Effects of inoculation of corn silage with *Lactobacillus hilgardii* and *Lactobacillus buchneri* on silage quality, aerobic stability, nutrient digestibility, and growth performance of growing beef cattle. *J. Anim. Sci.* 98:skaa267. <https://doi.org/10.1093/jas/skaa267>.
- NRC. 2001. Nutrient Requirements of Dairy Cattle. 7th rev. ed. Natl. Acad. Press.
- Oba, M., and M. S. Allen. 1999. Evaluation of the importance of the digestibility of neutral detergent fiber from forage: Effects on dry matter intake and milk yield of dairy cows. *J. Dairy Sci.* 82:589–596. [https://doi.org/10.3168/jds.S0022-0302\(99\)75271-9](https://doi.org/10.3168/jds.S0022-0302(99)75271-9).
- Oliveira, A. S., Z. G. Weinberg, I. M. Ogunade, A. A. P. Cervantes, K. G. Arriola, Y. Jiang, D. Kim, X. Li, M. C. M. Gonçalves, D. Vyas, and A. T. Adesogan. 2017. Meta-analysis of effects of inoculation with homofermentative and facultative heterofermentative lactic acid bacteria on silage fermentation, aerobic stability, and the performance of dairy cows. *J. Dairy Sci.* 100:4587–4603. <https://doi.org/10.3168/jds.2016-11815>.
- Oude Elferink, S. J. W. H., E. J. Krooneman, J. C. Gottschal, S. F. Spoelstra, F. Faber, and F. Driehuis. 2001. Anaerobic conversion of lactic acid to acetic acid and 1,2-propanediol by *Lactobacillus buchneri*. *Appl. Environ. Microbiol.* 67:125–132. <https://doi.org/10.1128/AEM.67.1.125-132.2001>.
- Peters, J., S. Combs, B. Hoskins, J. Jarman, J. Kovar, M. Watson, A. Wolf, and N. Wolf. 2003. Recommended methods of manure analysis. University of Wisconsin Cooperative Extension Publishing, Madison, WI.
- Queiroz, O. C. M., A. T. Adesogan, K. G. Arriola, and M. F. S. Queiroz. 2012. Effect of a dual-purpose inoculant on the quality and nutrient losses from corn silage produced in farm-scale silos. *J. Dairy Sci.* 95:3354–3362. <https://doi.org/10.3168/jds.2011-5207>.
- Reich, L. J., and L. Kung Jr. 2010. Effects of combining *Lactobacillus buchneri* 40788 with various lactic acid bacteria on the fermentation and aerobic stability of corn silage. *Anim. Feed Sci. Technol.* 159:105–109. <https://doi.org/10.1016/j.anifeedsci.2010.06.002>.
- Saylor, B. A., T. Fernandes, H. Sultana, A. Gallo, and L. F. Ferraretto. 2020. Influence of microbial inoculation and length of storage on fermentation profile, N fractions, and ruminal in situ starch disappearance of whole-plant corn silage. *Anim. Feed Sci. Technol.* 267:114557. <https://doi.org/10.1016/j.anifeedsci.2020.114557>.

- Souza, J. G., C. V. D. M. Ribeiro, and K. J. Harvatine. 2022. Meta-analysis of rumination behavior and its relationship with milk and milk fat production, rumen pH, and total-tract digestibility in lactating dairy cows. *J. Dairy Sci.* 105:188–200. <https://doi.org/10.3168/jds.2021-20535>.
- Sucu, E., and I. Filya. 2006. Effects of homofermentative lactic acid bacterial inoculants on the fermentation and aerobic stability characteristics of low dry matter corn silages matter. *Turk. J. Vet. Anim. Sci.* 30:83–88. <https://journals.tubitak.gov.tr/veterinary/vol30/iss1/13>.
- USDA-NASS. 2021. Corn Crop. Accessed Oct. 4, 2022. <https://quickstats.nass.usda.gov/results/337AA38F-4F03-3E5B-BB0D-7353E1D64A84>.
- Van Soest, P. J., J. B. Robertson, and B. A. Lewis. 1991. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* 74:3583–3597. [https://doi.org/10.3168/jds.S0022-0302\(91\)78551-2](https://doi.org/10.3168/jds.S0022-0302(91)78551-2).
- Wattiaux, M. A., and K. L. Karg. 2004. Protein level for alfalfa and corn silage-based diets: I. Lactational response and milk urea nitrogen. *J. Dairy Sci.* 87:3480–3491. [https://doi.org/10.3168/jds.S0022-0302\(04\)73483-9](https://doi.org/10.3168/jds.S0022-0302(04)73483-9).
- Weinberg, Z. G., O. Shatz, Y. Chen, E. Yosef, M. Nikbahat, D. Ben-Ghedalia, and J. Miron. 2007. Effect of lactic acid bacteria inoculants on in vitro digestibility of wheat and corn silages. *J. Dairy Sci.* 90:4754–4762. <https://doi.org/10.3168/jds.2007-0176>.
- White, R. R., M. B. Hall, J. L. Firkins, and P. J. Kononoff. 2017. Physically adjusted neutral detergent fiber system for lactating dairy cow rations. I: Deriving equations that identify factors that influence effectiveness of fiber. *J. Dairy Sci.* 100:9551–9568. <https://doi.org/10.3168/jds.2017-12765>.